Safety First — A Systematic Literature Review on the Impact of AI on High-Stakes Decision-Making

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Abstract

Abstract text.

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1. Introduction

The goal of this review is to explore safety-critical applications that have attempted or considered introducing artificial intelligence (AI) to their domains for decision-making purposes. We specifically exclude the use of AI for exploratory purposes to limit our scope to scenarios in which *human lives may be at risk*. While applications for the modern surge of AI-based technologies abound, we sought to identify the subset requiring the strictest scrutiny via its potential to cause the most harm. Our approach to identifying this subset was, as the title suggests, systematic. We cast a wide net in our search, then filter with a series of criteria, which we outline in Section 2. After distilling our initial pool of 1,135 papers into a final subset of 72 reports, we have identified critical features of both AI technology and implementation practices that span domain boundaries. Throughout this review, we explore patterns across these industries that demonstrate common failure modes, barriers to successful implementation, and limiting factors to status quo improvement with these technologies. The papers in this review range from software design and technical evaluations to legal reviews and qualitative studies with users and experts from the various domains. As such, while this paper was written by engineers, we anticipate its findings to interest regulators, practitioners, and participants of a variety of industries, alike.

2. Methods

This analysis follows the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) methodology and therefore collects documents in a systematic and automated way. To do so, we start by defining the research questions this review seeks to answer. These include, the titular: *What is the impact of AI on high-stakes decision-making?* As well as: *How are safety outcomes perturbed by AI-based decision-making in sensitive industries?* And *What are the safety risks of using AI to make decisions in sensitive industries?* While there are obvious benefits and harms to using AI-based technologies, that comparison alone is far too high-level to inspire targeted change within the AI technologies themselves. The nature of the questions guiding this review probe at what unique features and outcomes set AI-based decision-making mechanisms apart from traditional methods. Understanding these phenomena in detail would allow useful insight for both governmental agencies and engineers to mitigate evident shortcomings of this high-impact technology.

2.1. Query Term Compilation

After determining our research questions, we define a set of key terms using the Population, Intervention, Comparison, Outcome (PICO) methodology, and use the Cartesian product of these terms (one from each category) to determine relevant search queries. We use APIs for each of two databases, Scopus and Web of Science, to make these queries via a Python workflow, storing all resultant papers. Then, we remove duplicates based on digital object identifier (DOI), and merge the results of both database searches into a combined bibliography. Figure 1 shows visual representation of this workflow.

Critical to actually making these queries is defining the lists of keywords that comprise them. We start by defining our population terms.

We can organize the population terms we decided to include in this search in five major groups: nuclear and radiation, aviation, autonomous driving, energy and industrial processes, and medical technology and pharmaceuticals. These groups were selected for their highly sensitive nature, their propensity to use AI, and their regulatory authorities. This includes oversight by the U.S. Nuclear Regulatory Commission (NRC), the Federal Aviation Administration (FAA), the National Highway Traffic Safety Administration (NHTSA), the Occupational Health and Safety Administration (OSHA),

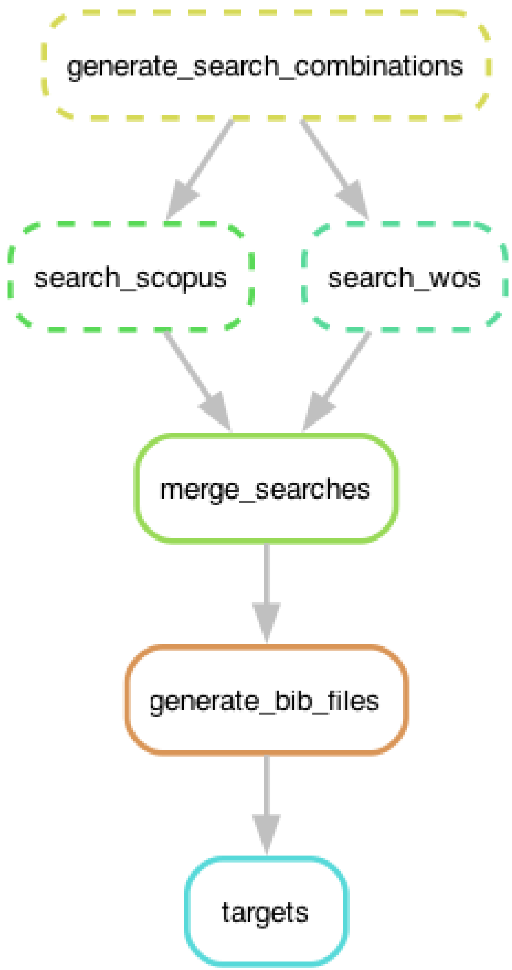

Figure 1: Directed acyclic graph showing automatic search workflow.

and the Food and Drug Administration (FDA). Obviously, there are a vast number of industries that might be considered "sensitive" and are federally regulated in the United States, however, due to time and space limitations, we prioritize those listed above in our analysis. While we included three query terms that did explicitly include some of these regulatory bodies (see Table 1), we found our more general terms sufficient to cover international bodies. Specifically, our goal was to avoid cases that consider risks *not* related to human health and safety (financial risks); we will discuss this further when we consider our exclusion criteria. The population search terms we include in our querying are shown in Table 1.

After determining our population terms, we then must consider our intervention terms. We determined this group by compiling a list of AI-related terms that would likely appear in any AI-based decision-making model. This includes words and phrases like "machine learning," "neural networks," and "language models." While this set surely does not contain every possible AI-related term, its goal is to target the major ones. We presume that almost any paper using AI will use the term "AI" itself, so the goal of additional terms is mostly out of precaution. In our full query, which we describe later in this section, we use the search operator `W/5` and `NEAR/5` for Scopus and Web of Science, respectively, to pair each intervention term in Table 1 with the wildcard term "decision*". In lieu of a mere `AND` operator, these operators require the word "decision" and its derivatives to be within five words of the AI-intervention term. The objective of this pairing is to return articles that use AI to either make or inform some kind of decision. The criticality and sensitivity of that decision will be evaluated during later screening. The intervention search terms we include in our querying are shown in Table 1.

Finally, we consider our outcome terms. Our research question focuses on the impact of AI-based decision-making on safety outcomes in sensitive industries. Thus, our outcome terms focus on safety, specifically, on risks to human health. We include general terms like "incident," "accident," and "near miss," as well as more industry-specific terms, like "misdiagnos*" and "operating experience," which cover medicine and power systems, respectively. To make this list extra comprehensive, and to ensure we capture any failures that led to official infractions, we included several regulation-specific terms too. This includes phrases like "FDA recall*" and "OSHA citation." A complete list of outcome terms used in our querying are shown in Table 1.

2.2. Search Process

The goal of our search process was to search two databases, Scopus and Web of Science, for all articles containing our four primary search terms in their title, abstract, or keywords. We imposed two additional requirements, that the papers be published after and not including 2014 and that the papers either be journal articles or conference papers from the top five AI conferences as listed by Google Scholar Metrics [1]. This conference paper subset includes *Neural Information Processing Systems*(NEURIPS), the *International Conference on Learning Representations* (ICLR), the *International Conference on Machine Learning* (ICML), the *AAAI Conference of Artificial Intelligence* (AAAI), and the *International Joint Conference on Artificial Intelligence* (IJCAI). We created this subset using the combination of both a paper type of `conference proceedings` and that conference proceeding containing one of the five acronyms listed above. In the case that this search catches an additional conference, that paper will be excluded during the abstract screening phase.

We combined our search terms into the following query search:

```
TITLE-ABS-KEY ( {population term} ) AND TITLE-ABS-KEY (
{intervention AI term} within five words of {decision*})
AND TITLE-ABS-KEY ( {outcome term} ) AND PUBLICATION YEAR
AFTER 2014 AND (DOCUMENT TYPE(article) OR (DOCUMENT TYPE(conference
proceeding) AND {conference subset}))
```

Note: the query format above matches Scopus syntax most closely, but is intended for demonstrative purposes only. We show our exact query structure in the code repository. The final set of articles considered for this analysis was pulled from both Scopus and Web of Science on December 15, 2025.

As shown in Figure 1, after searching the full set of queries in both Scopus and Web of Science, merging the search results into a single database and dropping duplicates based on DOI, we generated a singular bibliographic file in the `.bib` structure. We then loaded the bibliographic information into Zotero. Zotero flagged two articles as retracted, Mohanta et al. and Ponraj et al. We removed these records in the pre-screening phase along with one manually identified duplicate record. This yielded a total of 630 records for screening. Figure 2 shows the number of articles retained at each phase of the filtration process.

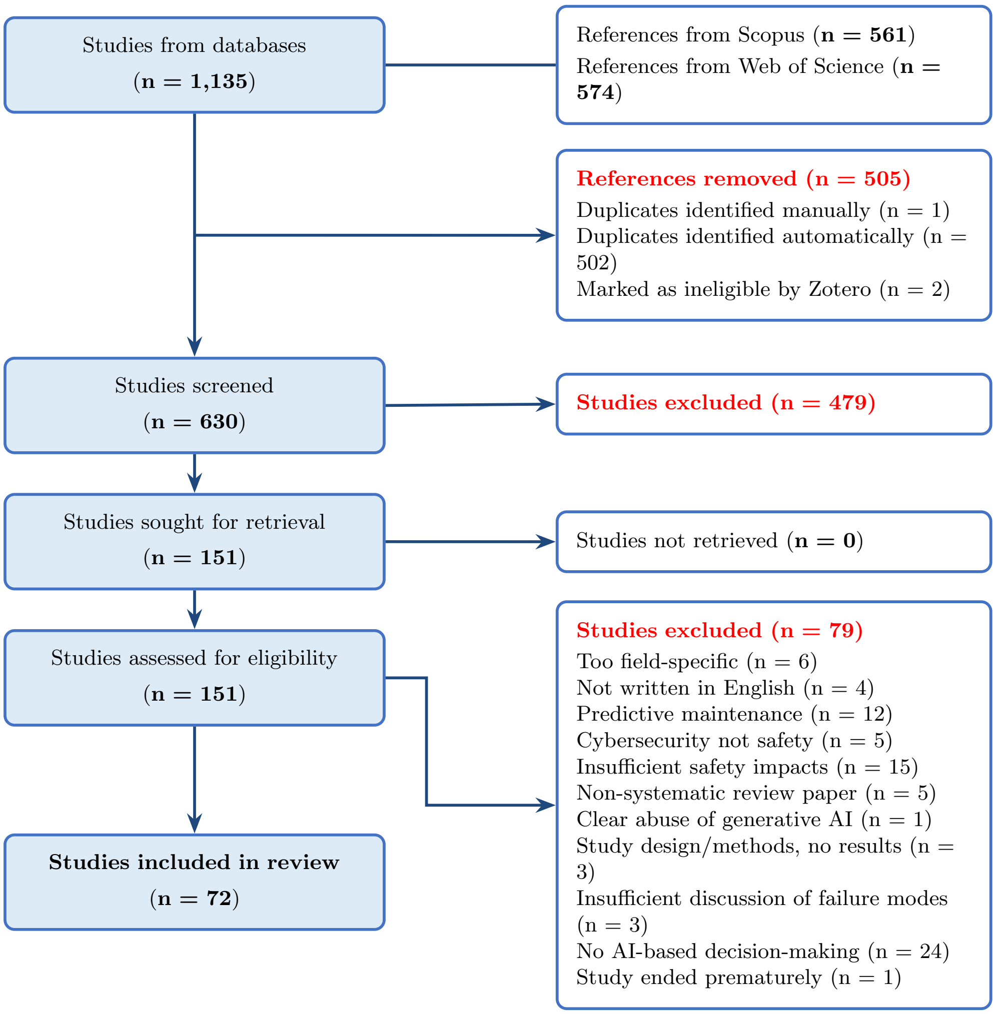

Figure 2: PRISMA flow diagram.

2.3. Screening Process

Following the PRISMA methodology, we perform the screening process in two phases– title and abstract screening, and full-text review. During the abstract screening, we followed the inclusion/exclusion criteria shown in Table 2, keeping questionable articles conservatively based on what we read in the titles and abstracts *only*. The purpose of this phase is to quickly remove all articles that absolutely do not meet our criteria, while keeping any that may. During the abstract screening, this mainly involved removing articles with an emphasis on outcomes that were not threats to human health or safety. We excluded many articles during this phase because they fell under the category of “predictive maintenance.” While predictive maintenance undoubtedly has value in a variety of industries, the goal of these types of studies almost always focuses on reducing costs associated with what would otherwise be regularly scheduled maintenance. Even in a seemingly safety-critical scenario– using AI to predict equipment failures in the intensive care unit (ICU), predictive maintenance only stands to improve the status quo of this system so long as it does not encourage *less* frequent maintenance. In these types of systems, there exists a conservative measure to ensure failures do not occur at a higher rate, which is to follow the maintenance schedule with the shortest intervals between checks, whether that be from an AI system or a traditional maintenance schedule. Because the safety-oriented solution in maintenance scenarios has a clear way to avoid harms caused by AI, and is fundamentally motivated as a cost-saving technique, we excluded predictive maintenance papers in our study unless they presented a clear safety-critical aspect that could not be mitigated by status quo processes. Likewise, we excluded all papers that sought to strictly reduce their system costs with AI or enhance their system security in a way that did not impact human health or safety. Our most basic heuristic: *Does this system have the potential to directly harm human lives?* was our main filtration system, however the additional details provided in Table 2 help provide more complete picture.

We also observed some exclusion criteria as to what constituted “AI.” Because we sought to consider regulatory approaches to AI in this analysis, we drew the line at pure linear regression based models and IF/ELSE-type logic models, since these algorithms are already widely accepted and regulated. Finally, we excluded any reviews that were non-systematic (this ended up including all review papers), any articles that were retracted, any ethics-based papers without a clear application to the AI technology or use of AI technology in their target domain, and any papers not written in English. To assist with our screening process, we used Covidence, a website designed for systematic reviews. We did not track reasons for exclusion during the abstract screening phase. As shown in Figure 2, during the abstract screening, we excluded 479 articles, yielding 151 articles for the full-text review process.

2.4. Full Text Screening

After the title and abstract screening, we proceeded to the full text screening stage. This phase, as its name suggests, involves reading the full text of each paper and applying the same inclusion/exclusion criteria shown in Table 2. The articles that made it past the abstract phase showed promise for meeting all of our inclusion criteria, but this must be confirmed after fully analyzing each text. During the full-text screening phase, we track the exclusion reason for each article. The number of articles and their specific exclusion criteria are shown in Figure 2.

We did include some additional exclusion reasons during this phase to increase our specificity. One example was the note: "Too field-specific." This was only a handful of papers, but consisted of studies that failed to discuss the AI technology in sufficient detail or conversely, failed to discuss the AI application in specific detail. Similarly, we did not include methods-only papers. One paper we reviewed showed clear abuse of generative AI in its creation with an apparent lack of editing or attribution, so it was excluded. Note, that we were able to access the abstracts of some articles in English, but not the full text, so two studies were excluded this way as well. Overall, we attributed the majority of exclusions to a lack of decision-making frameworks in the actual application of the AI. While we allowed for a variety of decision-making types (autonomous, augmented human-decision-making, human-in-the-loop, and even static benchmark setting), if the AI outputs were not applied for *any* type of decision-making, we excluded it. The goal of this was, again, to highlight the most safety critical scenarios that involve AI. If no AI-based decision-making takes place, it becomes challenging to attribute any of the outcome towards the AI. During the full-text review, we excluded 79 articles. The remaining 72 articles, as shown in Figure 2 is the final set included in this review.

2.5. Data Extraction

The final phase of our review methodology is the data extraction from the final subset of 72 articles. For this phase, we re-read each article in the final subset while taking notes of important findings and tabulating data for a variety of fields. These included: industry type, industry subcategory, AI type, comparison type, inclusion of explainability and type, consideration of regulation and regulatory authority, decision-making type, data type, whether the system was deployed, output feature considered, safety outcome, and country of origin. We determined these categories largely based on intuition from previous phases. Additionally, while the term "model explainability" has a variety of definitions, here we assume the most broad– anything that constitutes unpacking the logic of an AI model (interpretability, feature importance, etc.) counts.

During the data extraction phase, we also conducted a light quality assessment, in line with PRISMA guidelines. Due to the inhomogeneous nature of the types of papers included in this analysis, a single, premade quality assessment template would allow sufficient flexibility for our set of articles. Accordingly, instead of confirming if each article meets a certain set of criteria, we devised a set of "flags" to mark areas of concern for each article. Because this review considers a large portion of medical papers, we did evaluate each study for its consideration of bias, which is a known concern for AI applications in that field. To keep our assessment fair, we applied the same judgment for all other papers, which we believe is a reasonable assessment considering that all papers concern human health and safety and the way the AI views the differences between humans can therefore impact its judgment. Other than assessing for bias, we only flagged other categories if we had a concern while reviewing the articles. Commonly, this included flagging for limited dataset size. Most articles scored above 80 out of 100, so there were no major causes for concern regarding quality of the articles retrieved.

3. Quantitative Results

3.1. Querying Results

Figures 4a and 4b show quantitative querying results (the number of results per query combination) from Scopus and Web of Science as tree diagrams.

3.2. Data Extraction Results

The database used to tabulate our results can be accessed here. Additionally, we provide a database in Supplemental Materials.

After extracting the information mentioned in 2, we created a variety of figures to portray key trends from the set of articles reviewed. This includes country of author institution, main AI type considered in the article, decision-making type, whether regulation or explainability was considered, and types of explainability methods considered.

Figure 6 shows the number of articles that employed each type of AI in the final subset of articles. Note that many articles employed more than one

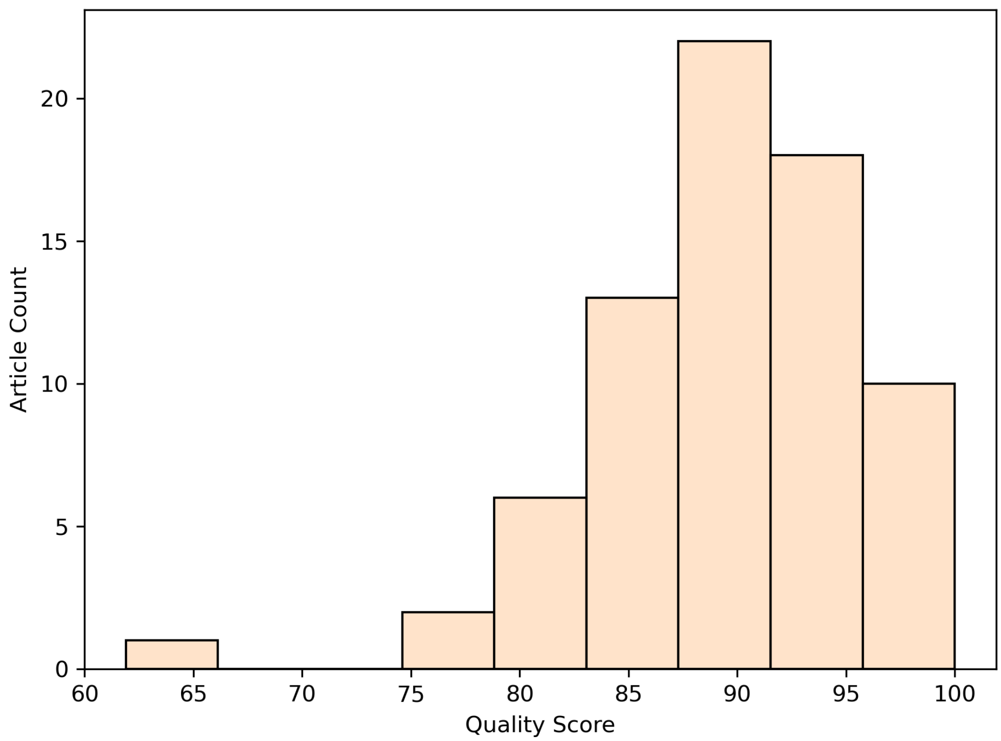

Figure 3: Quality assessment score distribution for articles in the final subset.

type of AI, so Figure 6 shows more counts than total articles. Similarly, some articles focused on ethical or regulatory frameworks and therefore considered AI more generally. We denote the AI type in these articles as "Non-Specific."

Figure 7 shows the decision-making type evaluated by the articles in the final subset. "Not system integrated" refers to articles that use AI to set some kind of threshold or benchmark value for decision-making, but do not actually integrate a dynamic version of the AI itself into the system under consideration. Additionally, we distinguish "augmented human decision-making" and "human in the loop" as systems that both require human action to make decisions, but human in the loop leans more heavily on the AI system whereas augmented simply uses the AI results as an additional piece of information to make a decision with. Autonomous systems do not necessarily require human oversight to take action, though they may have it for intervention purposes. This type of system is most typical with autonomous vehicles.

Approximately 49% of the articles in our final subset considered explanation methods in some capacity. Of those that did, Figure 8 shows which explanation method they employed. In some instances, authors mentioned the importance of explainability, therefore considering it, but did not conduct any explainability analysis themselves. These articles were denoted a "Non-Specific" explainability type in Figure 8.

Figures 9 and 10 show the number of articles who have at least one contributing author from the countries shown. Of the final subset of articles, approximately 44% considered regulatory oversight. This distribution of countries is therefore intended to show how the regulatory oversight bodies might be distributed and to highlight the complexity of defining AI regulatory frameworks that span international boundaries.

Finally, Table 3 shows the distribution of articles by industry for the final subset.

4. Qualitative Results and Discussion

The quantitative data provided in Section **??** only tell a fraction of the story. During the data extraction process, we took analysis notes and highlighted key findings from each study in the final subset. To best introduce these findings, we will first discuss each of the major industries identified in our study, highlighting key takeaways from the individual industries and some papers that brought especially unique points to our attention. Then, we will examine the similarities and differences between the various industries to determine what AI features and findings were universally beneficial, and which industries have specific challenges and demands.

4.1. Medical

Medical papers comprised nearly 60% of our final subset of articles (see Table 3). As the largest industry in article numbers from both the query results and the final subset analysis, the medical industry has a lot to offer in terms of experience using AI systems and highly sensitive usage demands. Due to its size, we break down this group of papers into smaller categories in the following subsections.

4.1.1. Pharmacology

Eight out of the 43 papers categorized as medical considered applications related to the prescribing and dispensing of medication— we categorized these papers under "pharmacology." Only two of these papers considered regulation, while three considered model explainability. Interestingly, two of the papers that did consider explainability evaluated large language models (LLMs), which are widely considered the most opaque.

These two articles [2] and [3] both evaluate LLMs as medication safety-focused clinical decision support systems (CDSS), including prescription errors and drug-drug interactions. Both [2] and [3] evaluated variants of Gemini, ChatGPT, and Claude. Ong et al. evaluate the success rate of identifying "drug-related problems" (DRPs) in three scenarios: pharmacist alone, pharmacist with AI-assistance (co-pilot mode), and AI alone and found co-pilot mode to yield the highest success of DRP identification [2]. Chase et al. do not consider AI-pharmacist cooperation, and instead evaluate the LLM independently, substantiating the finding that the LLMs perform suboptimally alone. They found the LLMs to make potentially harmful clinical recommendations up to 15.4% of the time (ChatGPT) [3]. Chase et al. also raise concerns for the less overtly dangerous but still problematic scenario of suboptimal recommendations, such as flagging non-clinically relevant drug interactions, which can lead to alert fatigue and subsequent oversights down the road [3].

In response to their findings, Chase et al. directly call for both external validation and benchmarking of medical AI against existing standards *and* model explainability and interpretability techniques to address the black-box nature of LLMs before they are implemented into clinical practice [3]. Conversely, Ong et al. state that the LLMs tested (ChatGPT, Gemini, and Claude) offer something that many "explainable AI tools" cannot— "conversational interfaces [that] allow clinicians to ask questions and receive context-specific explanations in plain language" [2]. While the conversational nature of chatbots may seem like a more approachable "explanation," this logic can be extremely dangerous as reasoning provided by a chatbot is *not* a true explanation and does not indicate *any* amount of model reliability, accuracy, or trustworthiness. We caution that this reasoning may lead to complacency and instead urge practitioners, regulators, and developers to consider genuine explainability and validation methods when applying these tools in practice.

The remaining articles in the pharmacology subcategory use either classical ML techniques (decision trees and gradient boosting) ([4], [5], [6]), association rule learning ([7]), and CDSS built-in systems ([8], [9]). Of these, Kieu et al. considered decision trees explicitly for their inherent interpretability [4]. From a deployment perspective, Liu et al. is a qualitative study of pharmacist interviews on trusting CDSS recommendations, which found that after consulting a CDSS, 78% of pharmacists did not adjust their dosing regimens due to "distrust and lack of understanding of the justification of the CDSS recommendations" among other concerns, despite a "generally positive attitude toward CDSS" [9]. This supports claims made by papers throughout this report that model trust is paramount to implementation and that this trust is often coupled with explainability or "justification." Even if a co-piloted pharmacist-AI approach yields the safest medication prescription and dispensing, without pharmacist trust in the partnership, the accuracy improvements are moot.

4.1.2. Imaging and Diagnostics

Eight of the 43 papers evaluated in this section evaluated using AI for diagnostic and/or imaging techniques. Of these papers, specific AI techniques included convolutional neural networks (CNNs) for classification of glaucoma [10], LLMs for emergency medical advice [11], generative adversarial networks (GANs) and other methods for skin lesion classification [12], decision trees for cervical spine injury diagnosis [13], and CNNs for stroke onset time prediction [14]. Of the remaining three papers, two included perspective papers from both leaders in emergency medicine on incorporating AI into their practices [15] and patients on AI being used to evaluate their prostate cancer screening MRI results [16]. These perspective studies will be evaluated with the other qualitative pieces in Subsection 4.1.5. The last paper in this category discusses AI in radiology more broadly and in light of the increased rate of AI use in published radiology papers, offers methods to increase generalizability by overcoming overfitting and underspecification [17].

Of the articles considered in this subsection, four of them consider regulation and five of them consider explainability. Of those that consider both, Vieira et al., who use various explainable AI techniques to support glaucoma diagnosis, presents one of the most explainability-forward papers in this review [10]. The authors use retinography images on CNNs coupled with CAM, GradCAM, LIME, SHAP, vanilla gradients, SmoothGrad, and SCIM to evaluate the quality of the different explainability methods for a high risk (blindness) clinical application [10]. The authors partnered with an ophthalmology specialist while conducting their analysis to evaluate the utility of the various explanations generated by the methods in their study. They found CAM-based methods to be the most effective way to reliably support diagnosis via its clear identification of clinically relevant regions and intuitive visual interpretation [10]. Vieira et al. present an extremely forward-thinking study in their work, anticipating and acting on the needs of clinicians to understand model results by evaluating a variety of explainability methods. While inherently interpretable models, like decision trees, can offer a clearer, more straightforward understanding of model logic, simpler models are not always an option. Especially in the imaging and diagnostic realm, the necessary data types (image-based) may not lend themselves well to the limited number of inherently interpretable ML methods available. In situations like these, to capitalize on more advanced model performance, we must develop ad-hoc explainability methods to understand the results in a way that makes them reliable and comprehensible for clinicians when the stakes are high. Vieira et al. do just that, and it is likely that future work will emulate and expand on their work in other highly sensitive domains as clinicians demand it and *especially* as regulation mandates it.

4.1.3. Radiation and Other Treatment

Nine of the 43 papers reviewed in this section evaluate the use of AI to design and carry out treatment plans with six of those nine specifically considering radiation treatment. The number of radiation treatment studies that employ AI alone represents the perceived potential utility of AI at improving highly sensitive treatment outcomes, as this subset of papers broadly discusses the precision required in radiation treatment. Of all nine papers considered in this section, six of them considered some aspect of explainability but only one of them considered regulation. The paper that considered regulation was an insulin dose optimization method for Type I diabetes already deployed in the AI-DSS DreaMed Diabetes Pro and considered oversight by the U.S. FDA but did not consider any explainability [8]. Of the radiation treatment papers considered, nearly all papers employed classical ML methods, including decision trees, random forests, support vector machines, and gradient boosting. Only one paper ([18])considered UNets, a type of CNN, exclusively, and another paper considered neural networks in addition to classical methods ([19]). Four ([20], [21], [19], [22]) of these six papers either conducted feature importance studies on these classical methods or rationalized using an inherently interpretable model for explainability purposes, therefore indicating a preference towards simpler, more comprehensible models for highly sensitive treatment methods. Most of these papers considered AI to find some threshold dose that is sufficient for treatment success while minimizing harm and secondary radiation side effects to the patient. One of the papers that did not consider explainability outright, did perform an uncertainty analysis on their UNet model for dose prediction to treat prostate cancer, which still highlights a need to establish model trust [18].

One paper in this section was particularly ahead of its time: Komorowski et al. used reinforcement learning to devise intensive care treatment strategies for sepsis, then use random forests to determine clinically interpretable reasoning (importance of model parameters) [23]. While not the most explicitly "explainability-focused," for a paper published in 2018, intensive care studies were already concerned with the black-box nature of their models and made attempts to extract model reasoning, knowing physicians would require it.

4.1.4. Legal and Regulatory

Eight of the 43 articles considered under the "medical" umbrella directly concerned legal and regulatory factors. Unlike other subsets, these papers generally do not evaluate a specific type of AI. Instead, they describe the ethics, liability, and broadly regulatory considerations that restrain these technologies in sensitive domains.

4.1.5. Qualitative Analyses and Roadmaps

4.2. Autonomous Vehicles

Autonomous vehicles comprised just over 18% of our final subset of articles, our second-largest category.

4.3. Nuclear

Nuclear systems comprised nearly 10% of our final subset of articles.

4.4. Aviation

Aviation comprised 8.33% of our final subset of articles (see Table 3). Of the aviation papers, three were specifically aimed at predicting accidents and incidents, two during flight [24], [25], and one during scheduling [26]. The three remaining studies looked at automated airport security detection [27], a Bayesian validation method for black-box systems that is currently employed by the FAA [28], and a roadmap for AI in flight advisory systems [29]. The airport security screening and the validation study are both deployed systems, while the rest of the studies present pure research. Of the six papers, only one, Cankaya et al., considered explainability [26]. They do so as a rationale for using Bayesian Belief Networks (BNNs). Unlike many scenarios typical of the medical literature, aviation AI targets low-probability, high-risk events. Cankaya et al. use this to justify needing interpretable models for the industry– creating a zero-accident ecosystem requires knowing the root causes of *potential* accidents before they happen. While "black-box" systems may provide insight on the frequency of failure, they cannot provide likely causes to enable prevention. Interpretable models may therefore fill this gap.

Other studies in this group [29], [25], [30], and [27] use AI to improve safety in various aspects of aviation (via flight advisory systems, hard landing warnings, engine flameout predictions, and accurate security screenings, respectively), but they do not evaluate model explainability or perform validation studies on their models. Such levels of model opacity and performance uncertainty leave their frameworks far from implementation from a regulatory perspective, except for [27], which evaluates an already deployed system that is separate from the mechanical functionality of the aircraft, and therefore subject to different scrutiny.

Notwithstanding their opacity, many non-interpretable models (such as those listed above) perform extremely well when tasked to predict an upcoming incident. While such models are obviously valuable, understanding their reliability, if not their exact inner-workings is critical for deployment. Moss et al. do just that in [28]. They model the failure probability estimation for autonomous flight systems using a Bayesian framework that "...iteratively fits a probabilistic surrogate model to efficiently predict failures" [28]. As the authors state, this system is actively involved in an FAA certification process. As such, other industries might apply a similar protocol to validate AI systems to their respective regulatory bodies. Over all industries, aviation contained one of the few validation studies (though other industries call for validation, such as in [31], which calls for "model auditing"). Though nearly all the industries considered in this review are regulated to some extent, the lack of validation studies, something we would typically expect for implementing new technologies into highly regulated fields, are evidently lacking.

It is not surprising, however, that the most rigorous validation study we found is from aviation, as the aviation industry is widely known for its safety-first culture. In fact, during our analysis, we found other industries explicitly reference aviation's success in implementing such a strong safety culture and their goal to replicate it. This is shown by Fuchs et al. in their paper "From Lindbergh's cockpit to predictive AI-driven health care: a roadmap for high-reliability medicine" [32]. Fuchs et al. emphasize the importance of validation studies like [28] when they state: "Regulatory agencies can formalize graded autonomy levels for medical AI, much like different categories of autopilot function, ensuring each stage undergoes thorough validation before broader adoption" [32]. They describe this as part of a larger roadmap to mimic aviation's historic approach to best safety practices, which relies on "robust governance"– a theme we observe throughout the papers in this review.

4.5. Other Industries

Combined, the remaining categories of ...

5. Conclusion

5.1. Study Limitations

1. Nearly all studies in this report found that AI technologies "improved outcomes" in some way. This is expected due to publication bias– null results simply do not get published with the same frequency as successful protocols. Second to this, of the 72 studies considered, only 21 involved what we considered a "deployed system" (meaning it was tested on real, critical scenarios). The remaining papers were either not deployed (38) or non-applicable as they were qualitative or roadmap papers (13). This means that even a broad systematic review like this cannot conclusively say if AI improves outcomes in sensitive industries. Future work might consider exclusively looking at deployed AI in a specific domain for more general conclusions on lives spared by AI, but even this effort will suffer the effects of publication bias. We also observe a chicken-and-egg problem in this facet—sensitive industries will be hesitant to deploy AI systems for safety-critical applications until they are proven safe and/or better than the status quo, yet the only way to evaluate this for a complete black-box technology is to study the outcomes of such a system on real scenarios. The solution is called for in papers throughout this review and explicitly demonstrated by Moss et al., a model-agnostic validation technique for black-box AI that can be reviewed by regulators [28]. That said, insufficient regulation renders the most effective validation techniques futile. Robust regulation must establish the validation criteria for model developers to apply. Failure to set the bar invites subpar technologies and risks human lives.

2. The authors of this review do not have expertise in all the fields evaluated. For this reason, there may be nuances in the subdomains that we missed. However, we believe the importance of cross-industry learning supersedes this potential limitation.

3. This review strictly sourced material from published academic articles on Web of Science and Scopus. As such, our review of relevant regulations for the variety of industries covered is necessarily incomplete. A review of AI-related regulations across domains and jurisdictions would certainly provide a more complete picture on the progress and consistency of the AI-legal landscape. Such a review is would likely require more legal expertise than the authors possess at this time. This in and of itself may underscore the disconnect between software developers, industry practitioners, and regulators. We urge each party to make efforts to close this gap to promote the safe use of this technology. Papers such as this serve as a first-step in that direction.

5.2. Future Work

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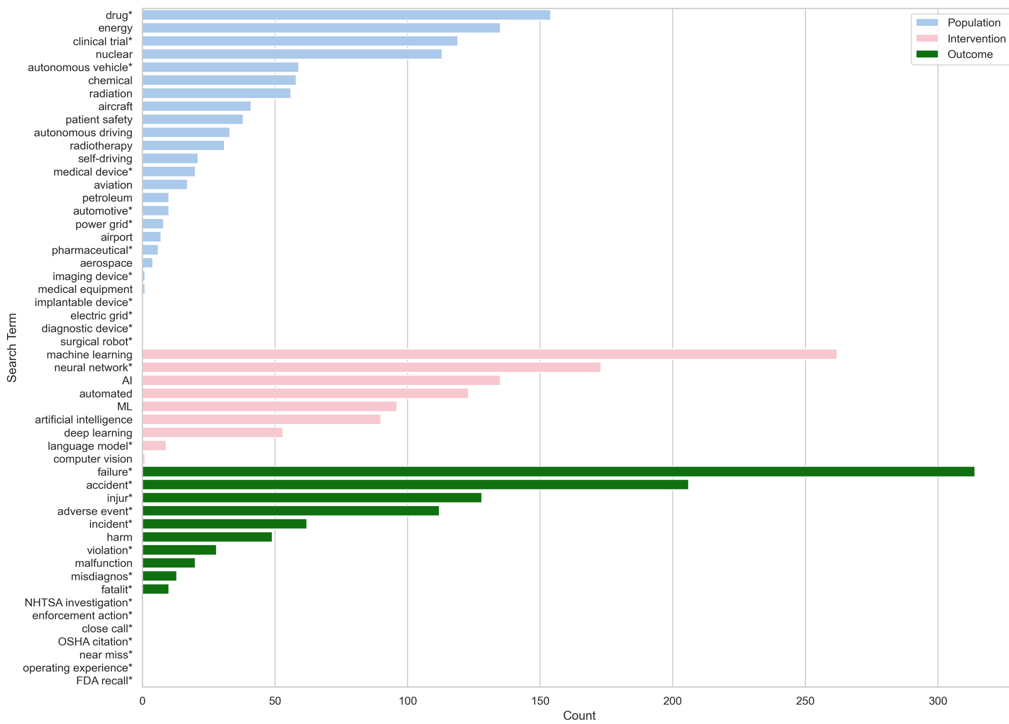
Table 1: Search terms used to produce Cartesian-product-based queries

Population Terms	**Intervention Terms**	**Outcome Terms**
nuclear	"artificial intelligence"	incident*
radiation	AI	"close call*"
aerospace	"machine learning"	accident*
aviation	ML	"near miss*"
aircraft	"neural network*"	injur*
airport	"language model*"	failure*
"autonomous driving"	computer vision	"adverse event*"
"autonomous vehicle*"	automated	malfunction
self-driving	"deep learning"	fatalit*
automotive*		misdiagnos*
energy		harm
"power grid*"		"operating experience*"
"electric grid*"		"OSHA citation*"
chemical		"FDA recall*"
petroleum		"NHTSA investigation*"
"medical device*"		violation*
"medical equipment"		"enforcement action*"
"diagnostic device*"		
"imaging device*"		
"surgical robot*"		
"implantable device*"		
drug*		
"patient safety"		
pharmaceutical*		
radiotherapy		
"clinical trial*"		

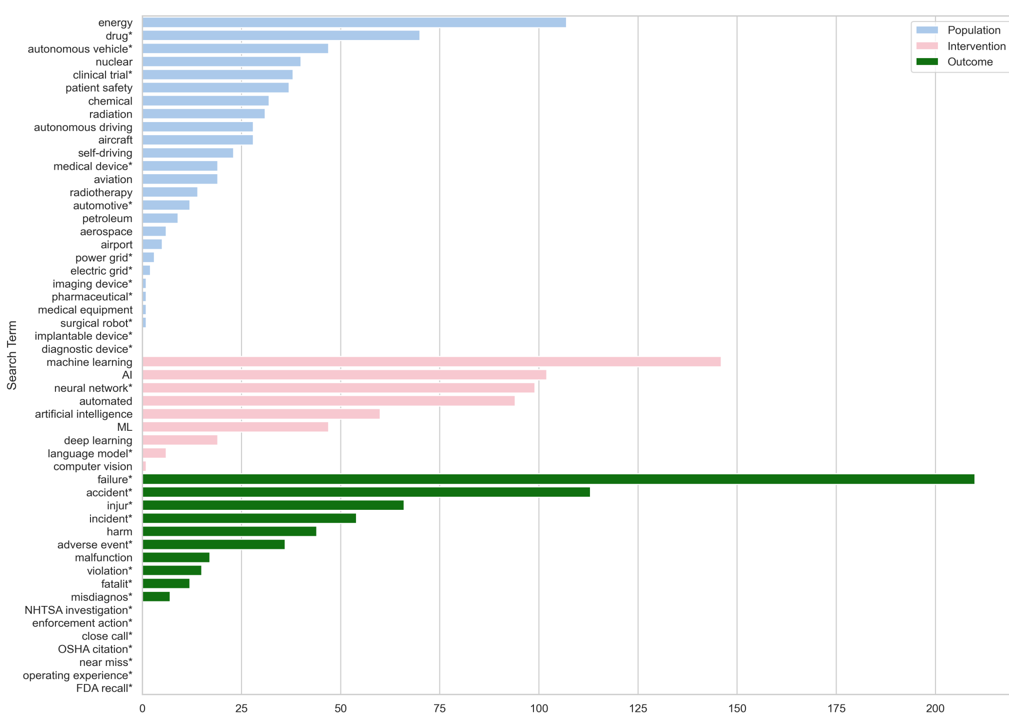
Note: Asterisks (*) indicate truncation wildcards.

	Inclusion	Exclusion
Population	Sensitive technology (e.g., nuclear, aviation) Medical studies, emphasizing equipment, devices, and drugs	Finance Human resources (even if health related) Cybersecurity
Intervention	AI-informed decision-making (with any degree of autonomy)	Simple flow chart or IF/ELSE logic Linear regression models
Comparison	Human evaluation Deterministic method	ML-ML comparison (with exceptions)
Outcome	Incidents, accidents, and failures Threat to human health Regulatory compliance violations	Non-threat to human health risks Financial risks or economic losses Cyber or other security risks
Other	Peer-reviewed journal articles Select conference papers	Non-systematic review papers Retracted articles Ethics without application Not written in English Published before 2015

Table 2: Inclusion and exclusion criteria used for abstract screening phase.



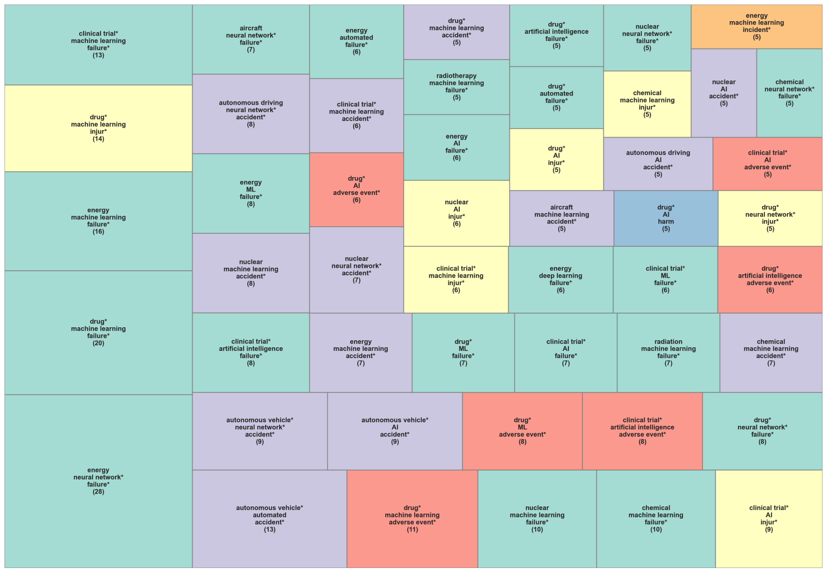
(a) Scopus



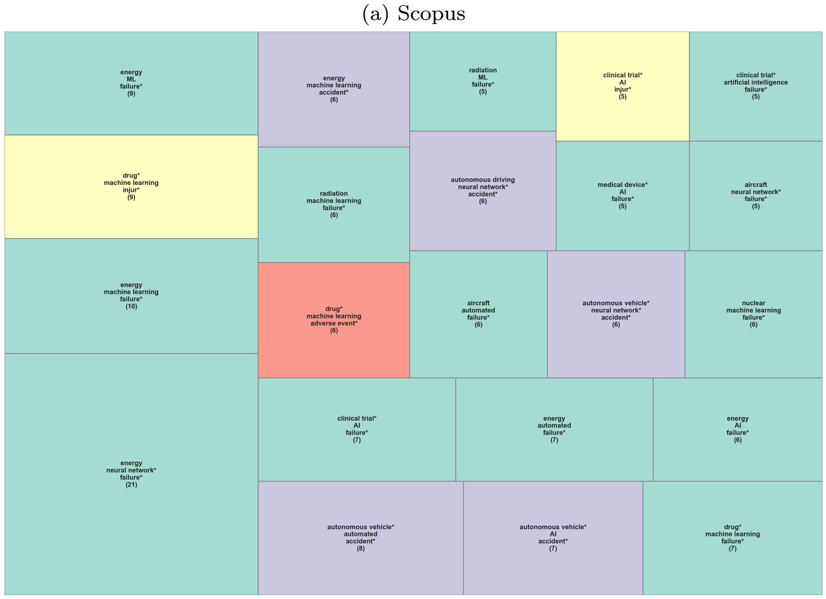
(b) Web of Science

Likewise, Figures 4 shows the sum of articles returned by each Scopus and Web of Science when individual search terms were included in all queries.

Figure 4: Sum of articles returned by each database when a given search term was included in all queries.



(a) Scopus



(b) Web of Science

Figure 5: Tree diagrams showing the sum of articles returned by each database for all query combinations that yielded > 4 results. Size of the box corresponds to the number of articles (also annotated) and color corresponds to outcome term.

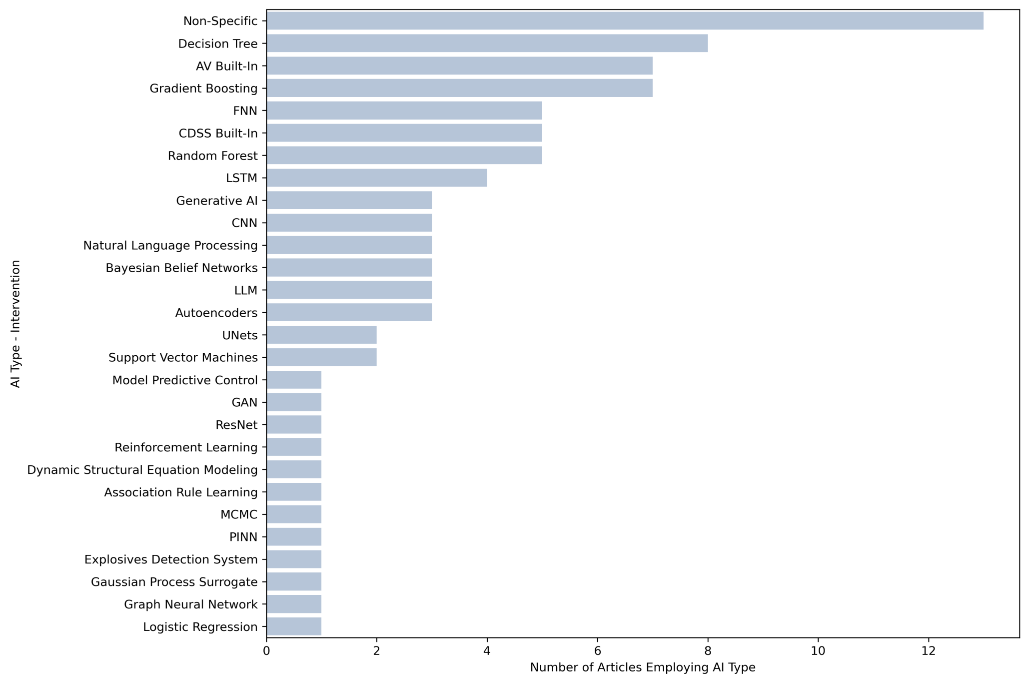


Figure 6: Number of articles in final subset that considered each type of AI method. Non-specific means the article did not provide details about the model type.

Industry	Percent of Final Subset of Articles
Medical	59.72%
Autonomous Vehicles	18.06%
Nuclear	9.72%
Aviation	8.33%
Ethics	2.78%
Manufacturing	1.39%

Table 3: Final article subset distribution by industry. Should I keep or remove the percent signs here?

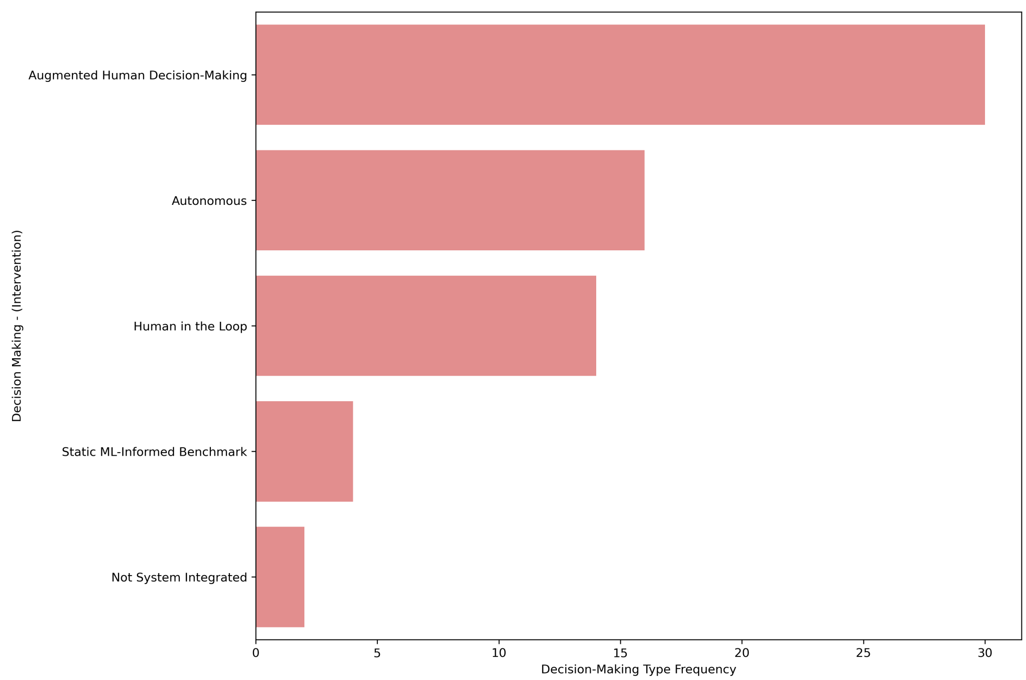


Figure 7: Decision-making types observed in final article subset.

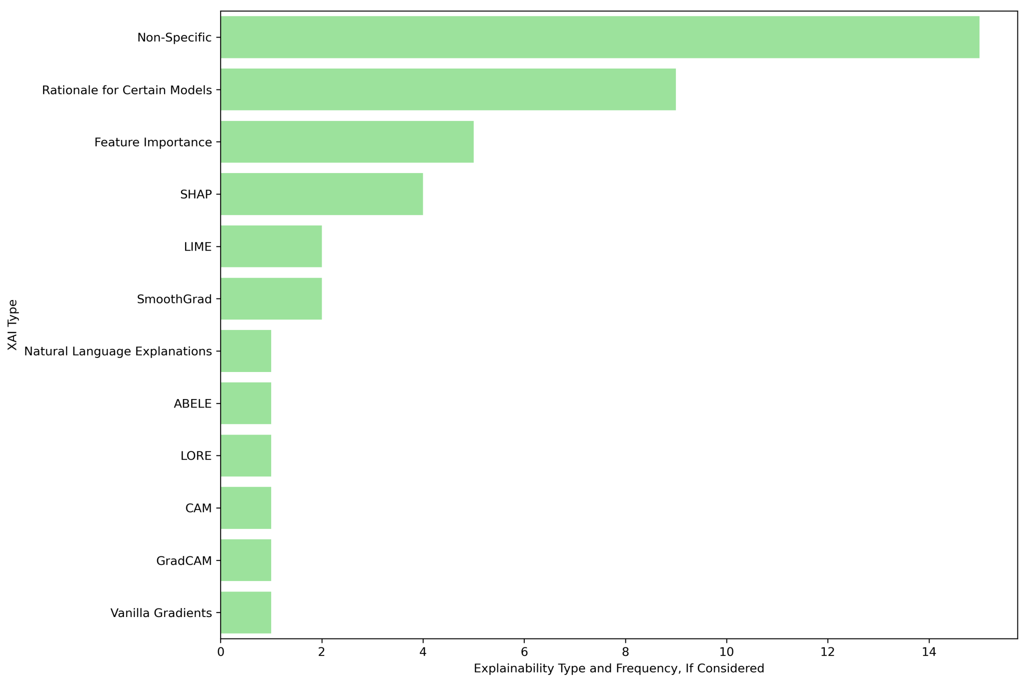


Figure 8: Explanation types observed in final article subset. “Non-Specific” indicates that explainability was considered but no method was applied.

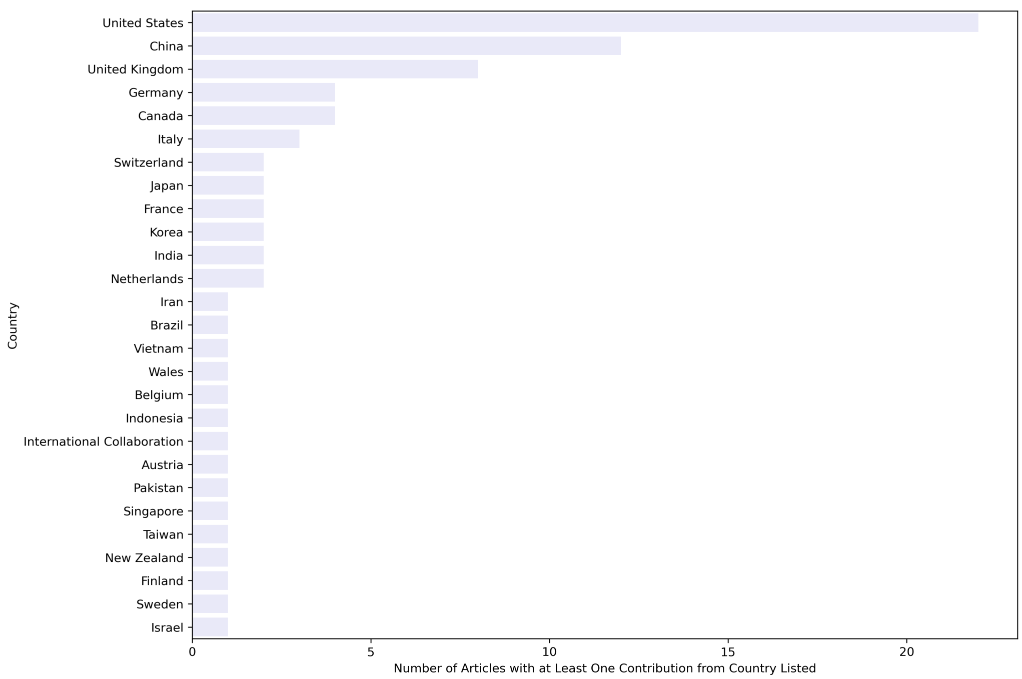


Figure 9: Number of articles in final subset who have at least one author from the listed country. "International Collaboration" denotes too many countries to list.

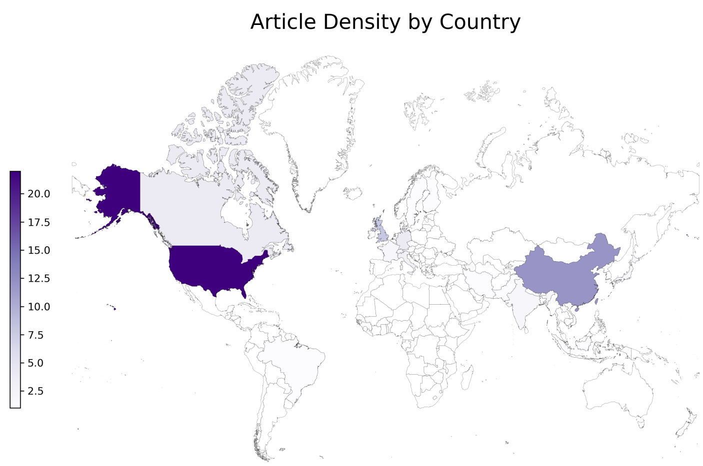


Figure 10: Number of articles in final subset who have at least one author from the listed country. "International Collaboration" article is excluded from this figure, see attached database for details.